

`jenkins.linuxgeek.in:8080`.

- c. Leave the URL field blank.
- d. Add read and write permissions to the repositories.
- e. Click the 'Save' button, and a key and secret will be automatically generated.

To ensure the Jenkins URL is set, take the following steps:

1. Open the 'Manage Jenkins' page.
2. Click on the 'Configure System' page.
3. Go to the section 'Jenkins Location'.
4. Set the correct URL to Jenkins' URL.
5. Click on the 'Save' button.

To add OAuth credentials to Jenkins, take the following steps:

1. Open the 'Manage Jenkins' page.
2. Click on 'Configure System'.
3. Go to the section 'Bitbucket Build Status Notifier plugin'.
4. If you still haven't stored the credentials, click 'Add', otherwise you can skip this step.
- a. Select a user name with password.
- b. Set the OAuth consumer key in the user name.
- c. Set the OAuth consumer secret in the password.
- d. Click the 'Add' button.
5. Select the desired credentials.
6. Click the 'Save' button.

Integrating Jira and Jenkins

To install the Jenkins plugin, take the following steps.

1. Go to the Jenkins dashboard:
Manage Jenkins -> Manage Plugins -> Available Plugins.
2. Search for Jira.
3. Select 'Atlassian Jira Software Cloud plugins' and click on 'Install'.
4. Restart Jenkins.

Create OAuth credentials in Jira for Jenkins, as follows.

1. Navigate to *Jira home > Jira settings > Apps*.
2. Select OAuth credentials.
3. Select 'Create credentials'.
4. Enter the following details:
 - a. *App name*: Jenkins
 - b. *App logo*: A URL to the Jenkins

logo, which will be used as an icon in the list of credentials; e.g., `http://jenkins.linuxgeek.in:8080`

- c. *Server base URL*: The URL to your Jenkins server; e.g., `http://jenkins.linuxgeek.in:8080`
- d. *Permissions*: Click on both deployments and builds
- e. Create credentials

Setting up Jenkins credentials

1. In Jenkins, go to the *Manage Jenkins > Configure System* screen and scroll to the 'Jira Software Cloud integration' section.
2. Select *Add Jira Cloud Site > Jira Cloud Site*. The site's name, client ID and secret fields are displayed.
3. Enter the following details:
 - a. *Site name*: This is the URL for your Jira Cloud site; e.g., `https://akshayithapedemo.atlassian.net/`.
 - b. *Client ID*: Copy from the OAuth credentials screen (Client ID column).
 - c. *Secret*: Select *Add > Jenkins*. For *Kind*, select *Secret* text. For *Secret*, copy from the OAuth credentials screen (Secret column). For *Description*, provide a helpful description.
4. Select 'Test connection' to make sure your credentials are valid for your Jira site.

For Jenkins multi-branch pipeline job configuration, take the following steps.

1. In Jenkins, click on 'New Item'.
2. Enter the job name and select 'Multi-branch Pipeline'. Click on 'Okay'.
3. In *General*, you give the name and description.
4. In *Branch Sources*, click on 'Git'.
 - a. Add the repository URL.
 - b. Add the credentials (user name and password of your Bitbucket account).
- c. *Behaviours* -> Select 'Discover branches'
- d. *Property strategy* -> Select 'All branches get the same properties'
5. In 'Build Configuration',
 - a. *Mode* -> Select 'By Jenkinsfile'

- b. Jenkinsfile name -> jenkinsfile
6. Just *Save and Apply*.
7. Click on 'Scan Multibranch Pipeline Log'.
8. We have run our job but it couldn't find jenkinsfile. So we create the 'jenkinsfile' and push it to our repository, jenkinsfile code as follows:

```
pipeline {
    agent any

    stages {
        stage('Build') {
            steps {
                bitbucketStatusNotify(buildState: 'INPROGRESS')
                echo "hello"
                bitbucketStatusNotify(buildState: 'SUCCESSFUL')
            }
            post {
                always {
                    jiraSendBuildInfo site: 'akshayithapedemo.atlassian.net'
                }
            }
        }
    }
}
```

9. Click on 'Scan Multibranch Pipeline Now'.
10. Click on 'Scan Multibranch Pipeline Log'.
11. Go to your Bitbucket, and click on 'Branches' where you will see that build status.
12. Go to Jira, open your issue and in the development section, you will get the build status. **END** 

References

- [1] <https://jenkins.io>
- [2] <https://www.atlassian.com/>

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